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Tran Ba Hoang

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Ba-Hoang Tran

Education

•Université Paris-Saclay *(suppose) September 2026 - July 2027*
Master 2 in Data, Knowledge and Hybrid Artificial Intelligence *Admitted, incoming*

•Vietnam National University - University of Engineering and Technology *September 2021 - June 2025*
Bachelor of Computer Science (Honors Program) *GPA: 3.61/4 (High distinction)*

- Graduation thesis: *Deep learning-based multimodal approaches for drug-drug interaction prediction.* Score: **9.7/10**.
- Recognized with the **Best Thesis Award** by Thesis Defense Committee.
- Relevant courses: Algebra, Calculus 1 & 2, Discrete Mathematics, Data Structures and Algorithms, Probabilities and Statistics, Object-oriented Programming, Information Theory, Machine Learning, Computer Vision, Natural Language Processing, Computer Network, Web Application Development, Software Project Management.

Experiences

•Vin Smart Future
AI Engineer *February 2026 - now*

- Engineered performance optimizations that reduced response time in VinFast's AI virtual assistant for real-time driver interactions (working in Speech & AI Agent components).
- Architecting system design and data workflows to optimize and deploy offline SLMs on VinFast's Qualcomm-based vehicle hardware, ensuring high-reliability performance without cloud dependency.

•Sun* Inc.
AI Engineer *June 2025 - Now*

AI-Powered Insight Automation project:

- Developed an automated AI system to collect, analyze, and report on around **1300 hours** of daily phone calls, delivering **same-day insights**.
- Architected a high-throughput inference pipeline by deploying Whisper on Triton Inference Server and GPT-OSS-20B via vLLM. Optimized pipeline performance through concurrent processing and load balancing, reducing processing time by **65%**.
- Leveraged prompt-tuning to automate the extraction of call reasons, sentiment classification, and concise summarization, transforming raw transcripts into structured business insights.

Sun* Assistant project:

- Developed a production-ready Slack bot integration to answer internal questions. Engineered an effective indexing system using advanced parsing & chunking mechanism and GraphQL, leveraging GraphRAG to map relationships across different technical documentation.
- Implemented AI Agents with persistent memory (Long-term/Short-term) stored via PostgreSQL to handle complex, multi-turn dialogues, resulting in a **70%** reduction in staff inquiries.

AI Research/Engineer Intern *November 2024 - May 2025*

- Developed a translation system using LLMs to accurately translate documents of various types (PDF, PPTX, DOCX, etc.) while preserving their original structure. Implemented a queue mechanism to efficiently manage concurrent requests, increasing processing speed by **60%**.

•Data Science and Knowledge Technology Laboratory - DS&KTLab - UET
Undergraduate Research Assistant *February 2023 - June 2025*

- Led research efforts on drug-drug interaction extraction using various methods, including NLP, knowledge graph, and multimodal approaches.
- Authored 3 peer-reviewed publications, including a paper in a **Q1 Journal** (*Journal of Biomedical Informatics*) and an accepted paper in the top-tier *ACM Multimedia* conference (**A***), focused on: 1) establishing a new Multimodal Biomedical Dataset (MUDI), and 2) proposing and analyzing diverse multimodal fusion strategies for Drug-Drug Interaction extraction.

•Tech Junior
Fullstack contractor *October 2024 - December 2024*

- Designed and implemented a full-stack web platform enabling teachers, parents, and students to purchase, organize, and manage online coding classes. Built and integrated front-end, back-end, and database components.

•Cinnamon AI
AI Researcher Intern *September 2023 - March 2024*

Robust Table Structure Detection & Recognition:

– Customized and fine-tuned new models to more accurately read and reconstruct tables, achieving a **27%** improvement for tables with partially missing borders.

Clinical Data Extraction from Unstructured Records:

– Collaborated with international clients to implement a practical AI solution for detecting and structuring data from diverse health check records with varying structures.

Trainee - Bootcamp member *June 2023 - August 2023*

– Selected as one of the four bootcamp members in Ha Noi for this competitive program with an acceptance rate of less than **10%**. Gained knowledge of software engineering best practices and machine learning algorithms through practice.

Publications

•Tung-Lam Ngo*, **Ba-Hoang Tran***, Duy-Cat Can*, Trung-Hieu Do, Oliver Y. Chén, and Hoang-Quynh Le (2025). "MUDI: A Multimodal Biomedical Dataset for Understanding Pharmacodynamic Drug-Drug Interactions". **Presented** in *33rd ACM International Conference on Multimedia*. In: arxiv:2506.01478 (*: Equally contributed.)

•**Ba-Hoang Tran**, Hung-Manh Hoang, Binh-Nguyen Nguyen, Hoang-Quynh Le, and Duy-Cat Can (2025). "Comprehensive Analysis of Multimodal Drug-Drug Interaction Extraction Using Diverse Fusion Strategies". **Published** in *Journal of Biomedical Informatics*. In: doi: 10.1016/j.jbi.2025.104874.

•Binh-Nguyen Nguyen*, **Ba-Hoang Tran***, Duy-Cat Can, Trung-Hieu Do, and Hoang-Quynh Le (2024). "Improving Drug-Drug Interaction Extraction from Biomedical Literature using Deep Multimodal Fusion". **Oral presentation** at *BME 2024*. In: doi:10.1007/978-3-031-90194-2_36. (*: Co-first authors.)

Achievements/Honors

•DAAD Study Scholarship for STEM Disciplines (Full-ride) *April 2026*

– One of the few Vietnamese recipients awarded a **full-ride Master's scholarship** by the German Academic Exchange Service (DAAD) for graduate study in Germany.

•Second prize of The Scientific Research Conference for Undergraduates 2025 at the University level *May 2025*

– Advanced from faculty-level first prize to the university-wide competition, earning **second prize** among top research projects across all faculties.

•First prize of The Scientific Research Conference for Undergraduates 2025 at UET-FIT *March 2025*

– Awarded the **first prize** at the annual VNU-UET IT Faculty undergraduate research conference, recognizing research quality, innovation, and presentation.

•Back2school KMS Scholarship Program for Outstanding Student *October 2024*

– Awarded a scholarship from KMS Technology company through a competitive selection process with an acceptance rate of **3.8%**.

•Top 2 University-Level Teams in the International Quant Championship 2024 *July 2024*

– Achieved **Top 2 university-level team** in the global quantitative finance competition by WorldQuant, with innovative financial modeling strategies demonstrating strong predictive power and robustness.

•Top 5 of Talent Acquisition Search Application Challenge on VPBank Hackathon Technology *May 2024*

– Worked on leveraging and customizing the Large Language Model pipeline with Retrieval Augmented Generation to rank resumes based on given job descriptions.

Leadership/Academic Activities

•Top 10 of Swin Hackathon 2024 *September 2024*

Team Leader/ AI Engineer

- This is a hackathon competition organized by Swinburne Vietnam University with 100+ teams registered.
- Developed a versatile system to generate short videos, documents, and slides from concepts using LLM, text-to-voice model, search capabilities, and RAG system, significantly cutting manual workload by **98%**.

•SOICT Summer School 2024 *August 2024*

Team Leader/ Participant

- Selected as a member of this **competitive, fully-funded** summer program to gain insights into trending AI topics, including Large Language Models and Diffusion.
- Leader of a 3-member team to make a project navigating the blind people powered by the multimodal language model.

•Top 12 regional of Vietnam GDSC Hackathon *January 2024*

Team Leader/ AI Engineer

- Created a prototype of an AI-powered tutoring system that assists students with homework and class material using text, images, and audio.

•FACT program between Singapore University of Technology and Design and VNU University of Engineering and Technology *July 2023*

Lecturer & Team Mentor

- Gave a lecture about basic deep learning concepts for first-year students of SUTD.
- Mentored a 3-member team to estimate the soil moisture from the sensor's data.

•UET Presentation Club *October 2021 - June 2024*

Organizer/ Member

- Core member of the Media department, responsible for creating propaganda campaigns and crafting social media posts that boosted reach by **10%** annually.
- Organizer of the Toi Ban Linh events in 2022 and 2024, significantly contributing to their success.

•National Conference on Breakthroughs in Science, Technology, Innovation and National Digital Transformation *January 2025*

Delegate

- One of 40 excellent students from VNU-UET to attend this **national-level conference**.

Personal Projects

•AI Voice Chat Assistant *January 2026*

Independent Developer

- Developed a low-latency conversational AI featuring real-time voice synthesis and interactive dialogue capabilities.
- Architected a multi-agent workflow utilizing LangGraph and MCP, integrating Superlinked for high-dimensional vector embeddings to enhance information retrieval.

•Laos-Viet Translation Model *December 2023*

Project Leader/ Developer

- Improved the BLEU score by **36%** from the baseline model by modifying the model's architecture and crawling publicly data.
- Acknowledged as the best Natural Language Processing class project.

•Image Generation using Generative Adversarial Network *November 2023*

Developer

- Re-implemented the models from the original GAN's paper. Applied advanced methods to reduce **20%** training epochs than other available similar architectures.
- Developed an enhanced GAN version that achieves Frechet Inception Distance by only **0.0285**, and conducted an ablation study.

•Image Classification Model Compression *July 2023*

Independent developer

- Optimized a model to be smaller than VGG16 by around **76%**, with only a **0.0586** increase in the top 5 errors by using quantization, pruning, and knowledge distillation.

•Flower Image Recognition Project *March 2023*

Independent developer

- Developed a flower image classification program using PyTorch, implementing transfer learning techniques from ResNet-50 and VGG16 models.
- Achieved an accuracy of **80%** after only 10 minutes of training on **102 Category Flower Dataset**. Received a certificate from Udacity as a reward for completion.

•Bomberman Game Project *November 2022*

Developer

- Made a game similar to Bomberman with improved interface and AI using Java. Creatively designed challenging mobs to enhance the game's difficulty.
- Achieved a score of **9.75/10** and ranked as one of the top **3** games in the class.

Academic Service

•Reviewer, Journal of Medical Artificial Intelligence (JMAI) *January 2026*

Notable Self-study Courses

•Neo4j Certified Professional *March 2025*

•Mathematics for Machine Learning and Data Science (Coursera) *June 2024*

•Generative AI with Large Language Models (Coursera) *May 2024*

•Deep Learning Specialization (Coursera) *May 2023*

•Machine Learning Specialization (Coursera) *March 2023*

Technical Skills and Interests

Programming/Markup Languages: **Python**, HTML, CSS, JavaScript, Java, C++, Bash scripting
Frameworks/Libraries: **Pytorch**, **Keras**, Matplotlib, Scikit-learn, Numpy, Pandas, LangChain, Neo4j, Postgresql, OpenCV, React, Tailwind, Node.js, Firebase, MongoDB, Flask, FastAPI, JavaFX, SDL2
Tools: Jupyter Notebook, Wandb, Neptune, GitHub, Git, Docker, Kubernetes, NGINX
Environments: GCP, AWS, Linux, Windows/WSL.
Areas of Interest: Information Extraction (RAG), Document Understanding, Multimodal Learning, Large Language Models, Natural Language Processing, Computer Vision
Languages: Vietnamese (native), English (C1 - IELTS 7.5)